

Profile

As a driven Computer Engineering student, I am deeply passionate about the intersection of robotics and artificial intelligence, particularly in the realm of embodied AI. My expertise spans robotics manipulation, computer vision, deep learning, large language models, and embedded systems. I am a proactive problem-solver with strong communication skills and a passion for continuous learning. Eager to embrace new challenges, I strive to integrate robotics and AI technologies to create impactful solutions that enhance human well-being.

WORK EXPERIENCE

Hong Kong Daimon Robotics Technology **Robotics Engineering Intern**

Shenzhen & Hong Kong, May 2025 — Aug 2025

- Developed and implemented CANOpen 301 and 402 protocols to enable seamless communication between control systems and motor modules for dexterous hand control and testing.
- Designed a Linux-based Qt application to streamline interaction between upper-level control systems and robotic hardware, enhancing testing efficiency.
- Implemented inverse kinematics for robotic arm control, enabling VR teleoperation in Isaac Gym and real-world deployment.

Belun Technology Company Limited **Embedded System Intern**

Hong Kong, Dec 2024 — Jan 2025

- Developed a test jig for medical device quality control, integrating STM32 microcontrollers, touchscreen displays, and peripherals for seamless communication.
- Designed PCBs with Altium Designer, and programmed OS-style firmware in C with IAR to enable communication via UART, QSPI, I2C, and USB protocols.
- Documented requirements, specifications, and testing procedures for project management.

Centre for Advances in Reliability and Safety **Technical Marketing Engineer Intern (Computer Vision)**

Hong Kong, Jul 2024 — Aug 2024

- Implemented, and fine-tuned object detection models leveraging the Swin Transformer backbone for detecting rail track surface defects.
- Co-authored a research paper proposing a novel integration method of Swin Transformer and CBAM, achieving a 7% improvement in mAP.
- Engineered and deployed automated marketing tools utilizing RAG LLM, LangChain, and Streamlit to streamline competition and funding searches.

RESEARCH EXPERIENCE

Robot Object Grasping and Placement (Team Lead, **HKUST Final Year Project)**

Aug 2025 — Present

- Developed and implemented an integrated framework to enhance dual-arm robot performance in coordinated object grasping and precise placement tasks.
- Performed extensive simulation studies across a range of tasks and object types

Robotic Hand Exoskeleton (Multi-I Robotic System **(MIRS) Lab), Undergraduate Research Opportunities** **Program (UROP)**

Feb 2025 — Jul 2025

- Created a robotic hand exoskeleton for adaptive gripping with pressure sensors and PID control.
- Programmed an STM32 microcontroller for servo control using AI voice recognition and a Wi-Fi-based web interface via UART and I2C protocols.
- Integrated computer vision by interfacing with the OpenMV module.

**Making Large Language Models Interact with
Physical World, Undergraduate Research Opportunities
Program (UROP),**

Jun 2024 — Jan 2025

- Collaborated with a Postdoc researcher to develop a novel LLM framework for an automated life journaling application utilizing phone sensor data.
- Created custom datasets for journal recording and Question Answering (QA) tasks.
- Evaluated the LLM framework for automated life journal generation with NLP.

EDUCATION AND COURSES

Hong Kong University of Science and Technology
Bachelor of Computer Engineering

Sept 2022 — Jun 2026

- **Dean's List**, School of Engineering (Spring 2023-2024, 2024-2025 | TGA > 3.9)
- **HKDSE Scholarship | HKWPEA Charitable Foundation WISE Scholarship**
- **Relevant Coursework:** Artificial Intelligence for Medical Image Analysis (A+), Introduction to Mobile Robotics (A+), Introduction to Embedded Systems (A+), Deep Learning in Computer Vision (A), Reinforcement Learning (A)

University of Oxford (Exchange)
LMH Oxford Summer Programmes (Advanced Artificial
Intelligence and Machine Learning: Reinforcement
Learning and Computer Vision)

Jul 2025 — Aug 2025

- 1st Class Honor

NVIDIA Deep Learning Institute
Fundamentals of Deep Learning

Feb 2025

DeepLearning.AI and Stanford University (Coursera)
Machine Learning Specialization

Jan 2024

PUBLICATIONS

**CBAM-SwinT-BL: Small Rail Surface Defect Detection
Method Based on Swin Transformer with Block Level
CBAM Enhancement**

Dec 2024

Jiayi Zhao, Alison Wun-Lam Yeung, Muhammad Ali, Songjiang Lai, Vincent To-Yee Ng IEEE Access, Dec 2024

COMPETITIONS

**Joint University Eco-Business Innovation Award
2022-2023 - Champion (Leader) & Best Presenter**

Apr 2023

**HKUST Sustainable Smart Campus Student
Competition 2022 - Campus Project Award & Most
Innovative Solution Award**

Dec 2022

Open GIS Competition 2022 - Champion (Leader)

Jul 2022

**IFTA Girls in Tech Hackathon 2021 - Second
Runner-up (Leader)**

Aug 2021

COMMUNITY INVOLVEMENT

HKUST Undergraduate Student-Initiated Experiential Learning (USEL) Lab - Student Helper

Sept 2023 — Dec 2023

- Guided students in hands-on projects, including Arduino circuit design, 3D printing, laser cutting, power tool operation, and drone piloting.

Campus Project "Paper to Infinity" at HKUST - Organizer

Dec 2022 — Oct 2023

- Organized workshops for students to create keychains from recycled paper using 3D-printed molds, achieving 100% participant satisfaction.

HKUST - The Next Changemaker Program - Project Lead

Sept 2022 — Apr 2023

- Planned a community service project addressing the animal abandonment issue in Hong Kong.

TECHNICAL SKILLS

Programming	Python, C++, C, TypeScript
AI/ML & LLM	Machine Learning, Deep Learning, Reinforcement Learning, Frameworks: PyTorch, TensorFlow, LLM Applications
Robotics	Isaac Gym, Pinocchio, Curobo
Embedded Systems	STM32, ESP32, Arduino
Mobile & Full-Stack Development	React Native (TypeScript + JSX), Supabase (PostgreSQL)

LANGUAGES

Cantonese (Native), Mandarin (Fluent), English (Fluent)

Links

[Personal Website and Portfolio](#)